Where the following statement is normally added?

w.mainloop( )

Select one:

- First line of GUI code
- Last line of GUI code
- Second line of GUI code

Which of the following convert the image from pixel based to feature based?

Select one:

- backend of CNN
- frontend of CNN
- Deep learning
- RNN
- hidden layers

Read the following line of code.

```
myObj = ImageDataGenerator ( brightness_range=(0.3, 0.7) )
```

What is the target of this line?

Select one:

- Generate new set of brighter images than the original one.
- Generate new set of images that represent zoomed-in images of the original one (seems to be bigger).
- Generate new set of images that represent zoomed-out images of the original one (seems to be smaller).
- Generate new set of darker images than the original one.

In the speech recognition applications, the input is normally a ..... and the output is normally a .....

Select one:

- English Chinese
- audio text transcript
- text transcript audio
- audio audio+video
- text audio

Which of the following statements is correct?

Select one:

- In recent years, the challenge tasks in ImageNet competition include labeling videos tasks.
- The object detection challenges in the ImageNet competion require prediction the classes of objects present in an image without drawing a bounding box.
- All of the ImageNet images have bounding box annotations.
- There are more than 100 million images in the ImageNet dataset.

Read the following segment of code.

```
model = Sequential ()

model . add ( Conv2D ( 12 , kernel_size =( 3 , 3 ),
strides=(2,2), padding = "same" ,input_shape =(120,120,1)))

model . add ( ReLU ())

model . add ( MaxPooling2D ( pool_size =(2,2), strides =(2,2)))

model . add ( Flatten ())
```

How many elements are placed in the 1D vector after the flattening process?

Select one:

- 10,800
- 120
- 36
- 5,400
- 21,600
- 1200
- 1024
- 43,600

Read the following segment of code.

```
model = Sequential ()

model . add ( Conv2D ( 16 , kernel_size =( 3 , 3 ),strides=(2,2),
padding = "same" ,input_shape =( 200 , 200 , 1 )))

model . add ( ReLU ())

model . add ( MaxPooling2D ( pool_size =(4,4), strides =(4,4)))
```

What is the size of each pooled map?

Select one:

- 50 x 50
- 16 x 200
- 100 x 100
- 16 x 16
- 25 x 25

Read the following segment of code and fill in the blank to create a window.

```
import tkinter as tk

myWindow = ……..
```

Select one:

- w.title('Hi')
- tk.grid(row = 0)
- tk.title('Hi')
- tk.Tk( )

Read the following segment of code.

```
import os
def loadDirectory ( parent ):
    numOfsomething = 0
    for subdir in os . scandir ( parent ):
        if not subdir . is_dir ():
            print ("Okay")
            continue
        numOfsomething = numOfsomething + 1

loadDirectory ( 'collection2' )
```

Suppose that there are 3 images and 6 folders placed in "collection2".

Which one of the following statements is correct?

Select one:

- "Okay" will be printed 0 times, and the value of "numOfsomething" will be 3
- "Okay" will be printed 3 times, and the value of "numOfsomething" will be 0
- "Okay" will be printed 3 times, and the value of "numOfsomething" will be 6
- "Okay" will be printed 6 times, and the value of "numOfsomething" will be 6
- "Okay" will be printed 6 times, and the value of "numOfsomething" will be 3

Suppose that you have a data set of 60 rows, 20 columns.

Which one of the following statements is logically not acceptable?

Select one:

- model.fit(X_train, Y_train, epochs=50, batch_size=15)
- model.fit(X_train, Y_train, epochs=70, batch_size=30)
- model.fit(X_train, Y_train, epochs=80, batch_size=10)
- model.fit(X_train, Y_train, epochs=150, batch_size=80)
- model.fit(X_train, Y_train, epochs=100, batch_size=20)
- model.fit(X_train, Y_train, epochs=300, batch_size=12)

Fill in the blank.

The command …… is for one-line text, but the command …… is for multiline text.

Select one:

- Text Entry
- Radiobutton Checkbutton
- Entry Text
- Checkbutton Radiobutton

Which one of the following simulation tools is used by Mitsubishi?

Select one:

- Gazebo
- Workspace
- Webots
- MRDS
- V-REP

Fill in the blank.

Hidden layers are normally used in neural networks when there is ..... relationship.

Select one:

- linear
- nonlinear

You are writing a Python code to display 12 images in 3 rows and 4 columns as below.

What is the correct line of code to display the image pointed to in the figure above?

Select one:

- pyplot.subplot(346)
- pyplot.subplot(436)
- pyplot.subplot(3, 4)
- pyplot.subplot(12, 2)
- pyplot.subplot(345)

Read the following segment of code:

```
import cv2 , imutils
myImg = cv2 . imread ( 'street.jpg' )
Img2 = imutils . rotate_bound ( myImg , 20 )
```

which of the folowing statements is correct?

Select one:

- Img2 should have the same width and height as myImg
- The method rotate_bound does not accept the number 20 as a second parameter. you should use just 90, 180, or 270.
- Img2 and myImg may have different width and height
- The width of myImg will be the same as the height of Img2, and the height of myImg will be the same as the width of Img2.

Look at the following 3 steps.

step#1: using (asarray) method

step#2: using (astype) method

step#3: dividing pixel values by 255.0

The target of the steps above is to .....

Select one:

- remove digital noise from images
- decrease the time of the training
- crop the bounds of images
- apply deep learning with 255 hidden layers
- clean your data from missing values

Read the following segment of code.

```
import cv2
import numpy as np
myImg = cv2 . imread ( 'sun.jpg' )
m = myImg . shape [ 0 ]
image2 = np . roll ( myImg , m // 20, axis=0)
```

What does the previous code do?

Select one:

- shift the image left by 20%
- shift the image left by 20 pixel
- shift the image right by 20 pixel
- shift the image down by 20%
- shift the image up by 20 pixel
- shift the image up by 20%
- shift the image right by 20%

Which one of the following statement is correct?

Select one:

- In the supervised learning, the test data is used to derive a model (find a relation between inputs and outputs), and the training data is used to evaluate the accuracy.
- In the supervised learning, the training data and the test data are both used to derive a model (find a relation between inputs and outputs).
- In the supervised learning, the training data is used to derive a model (find a relation between inputs and outputs), and the test data is not used for any purpose.
- In the supervised learning, the training data is used to derive a model (find a relation between inputs and outputs), and the test data is used to evaluate the accuracy.

Read the following segment of code.

```
model . add ( Conv2D ( 24 , kernel_size =( 3 , 3 ), strides=(2,2), padding = "same" , input_shape =( 100 , 100 , 1 )))
```

How many feature map will be generated?

Select one:

- 9
- 100
- 10,000
- 24
- 240,000

Fill in the blank:

All of the following operations can be applied using "transpose" method except .....

Select one:

- shift to right 10%
- rotate 180 degree
- flip on the horizontal axis